INCH-POUND

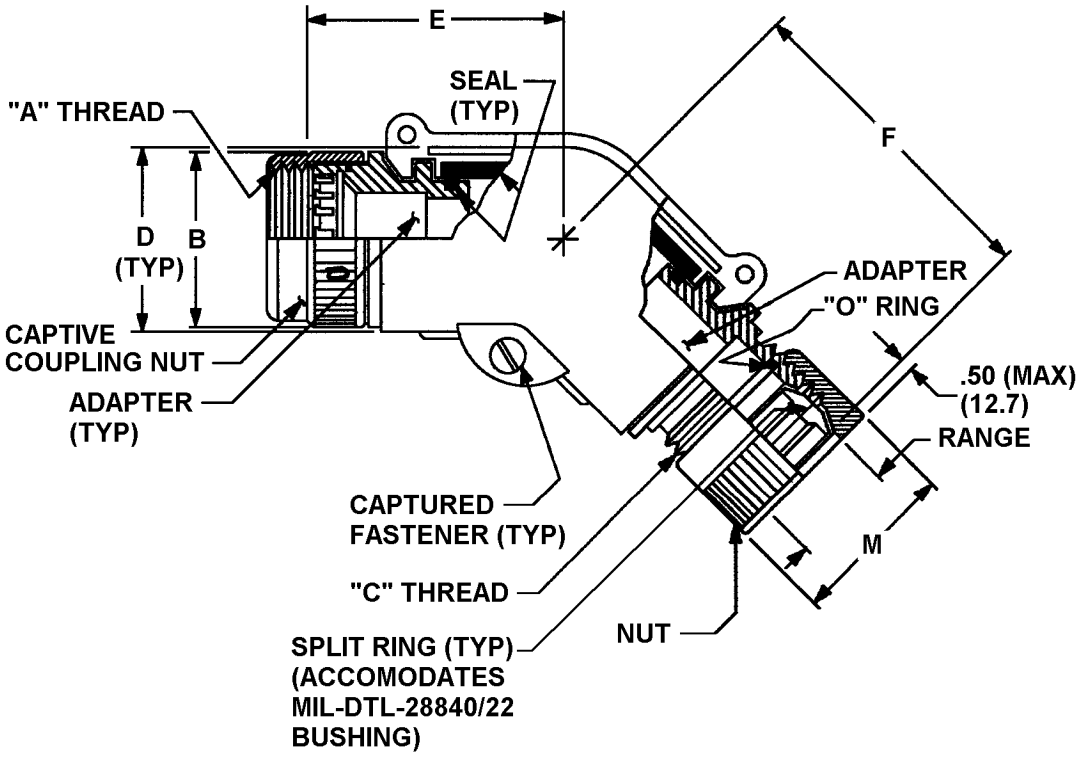
MIL-DTL-28840/27C
11 January 2010
SUPERSEDING
MIL- DTL -28840/27B
20 April 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 45°, METAL CONDUIT FOR EMI SHIELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.



STYLE A

FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5935

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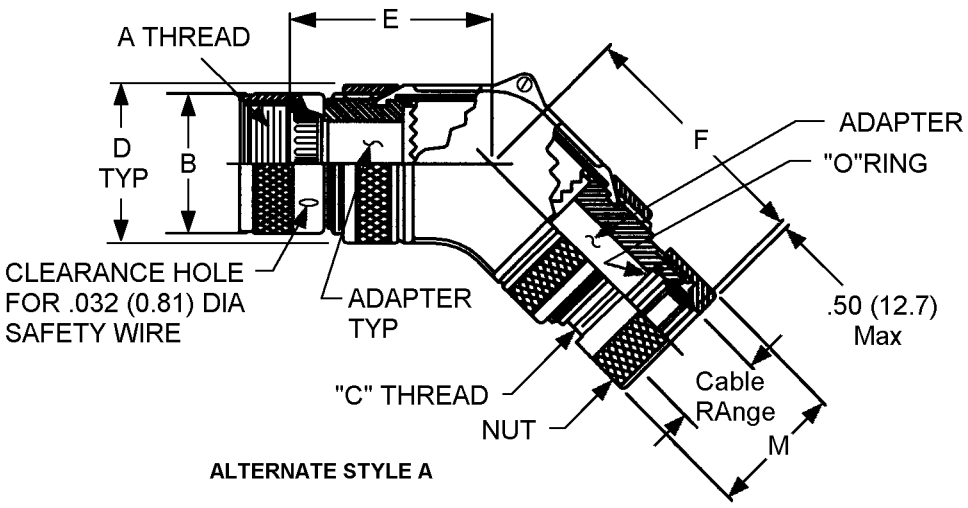
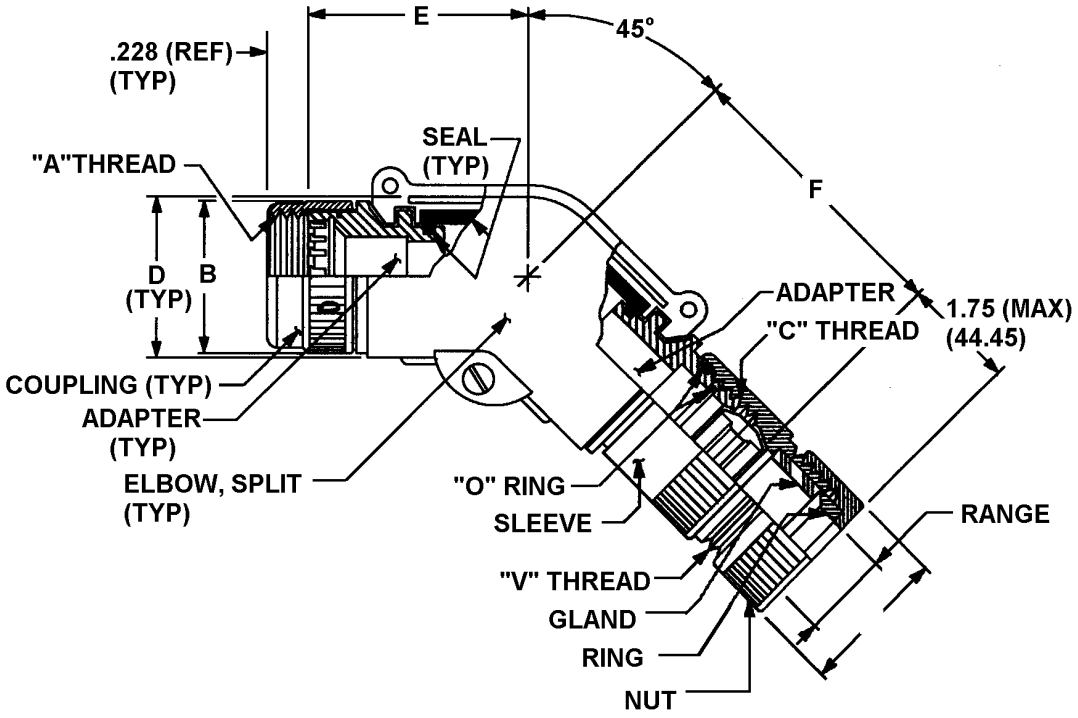


FIGURE 1. Dimensions and configurations - Continued.

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STYLE B

FIGURE 1. Dimensions and configurations - Continued.

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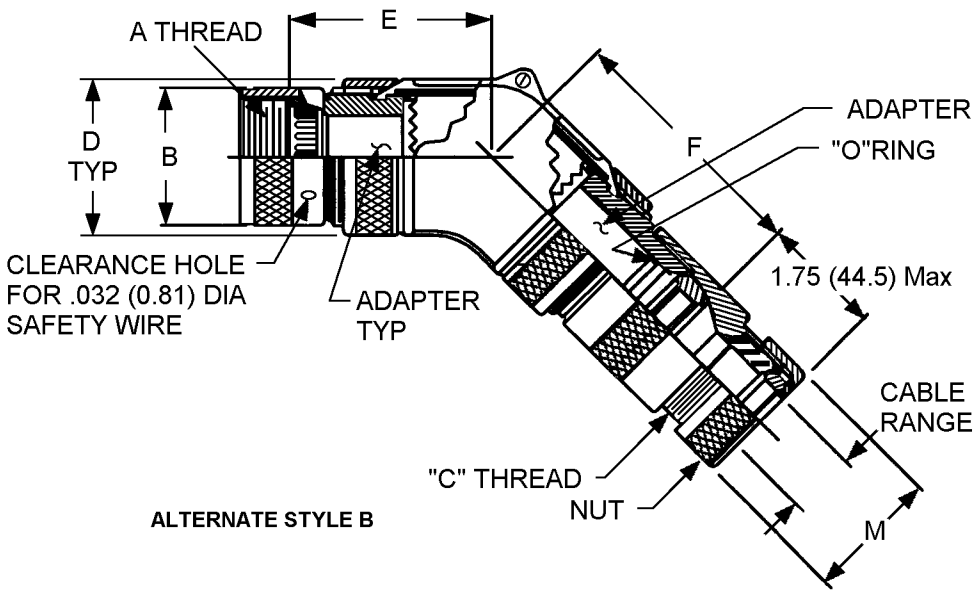


FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
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Dash no.	Designator (see note 1)	A thread class 2B	B max dia	C thread class 2
1	A(11)	.750-20UNEF	1.028 (16.11)	.875-20UNEF
2	B(13)	.875-20UNEF	1.141 (28.98)	.875-20UNEF
3				1.000-20UNEF
4	C(15)	1.000-20UNEF	1.263 (32.08)	.875-20UNEF
5				1.000-20UNEF
6				1.125-18UNEF
7				1.250-18UNEF
8	D (17)	1.125-18UNEF	1.387 (38.23)	1.125-18UNEF
9				1.250-18UNEF
10				1.625-18UNEF
11	E (19)	1.250-18UNEF	1.513 (38.43)	1.125-18UNEF
12				1.250-18UNEF
13				1.625-18UNEF
14	F (23)	1.438-18UNEF	1.703 (45.51)	1.250-18UNEF
15				1.625-18UNEF
16				1.875-18UNS
17	G (25)	1.562-18UNEF	1.825 (46.35)	1.250-18UNEF
18				1.625-18UNEF
19				1.875-18UNS
20				2.125-16UN
21	H (29)	1.875-16UN	2.143 (54.43)	1.875-18UNS
22				2.125-16UN
23	J (33)	2.062-16UNS	2.329 (50.16)	2-125-16UN
24				2.750-16UN
25	A (11)	.750-20UNEF	1.028 (16.11)	.875-20UNEF

FIGURE 1. Dimensions and configurations - Continued.

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Dash no.	Designator (see note 1)	D max dia	Sealing range		Conduit size MIL-DTL-28840/4	E max dim
			Max	Min		
1	1(11)	1.031 (26.19)	.63 (16.0)	.50 (12.7)	03	2.36 (59.9)
2	B(13)		.63 (16.0)	.50 (12.7)	03	
3			.75 (19.1)	.63 (16.0)	04	
4	C(15)	1.281 (32.50)	.63 (16.0)	.50 (12.7)	03	2.56 (65.0)
5			.75 (19.1)	.63 (16.0)	04	
6			.88 (22.4)	.75 (19.1)	05	
7		1.531 (38.80)	1.00 (25.4)	.88 (22.4)	06	2.67 (67.8)
8	D (17)		.88 (22.4)	.75 (19.1)	05	
9			1.00 (25.4)	.88 (22.4)	06	
10			1.38 (35.1)	1.19 (22)	08	
11	E (19)		.88 (22.4)	.75 (19.1)	05	
12			1.00 (25.4)	.88 (22.4)	06	
13		1.781 (45.24)	1.38 (35.1)	1.19 (30.2)	08	3.05 (77.5)
14	F (23)		1.00 (25.4)	.88 (22.4)	06	
15			1.38 (35.1)	1.19 (30.2)	08	
16			1.63 (41.4)	1.44 (36.6)	10	
17	G (25)		1.00 (25.4)	.88 (22.4)	06	
18			1.38 (35.1)	1.19 (30.2)	08	
19			1.43 (41.4)	1.44 (36.6)	10	
20		2.031 (51.59)	1.88 (47.8)	1.63 (41.4)	12	3.33 (84.6)
21	H (29)		1.63 (41.4)	1.44 (36.6)	10	
22			1.88 (47.8)	1.63 (41.4)	12	
23	J (33)	2.531 (64.29)	1.88 (47.8)	1.63 (41.4)	12	3.70 (94.0)
24			2.38 (60.5)	2.13 (54.1)	16	
25	A (11)	1.031 (26.19)	.75 (19.1)	.63 (16.0)	04	2.36 (59.9)

FIGURE 1. Dimensions and configurations - Continued.

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Dash no.	Designator (see note 1)	F max dim	V thread class 2	M max dia	Y max dia
1	A (11)	3.50 (92.8)	1.000-20UNEF	1.12 (28.5)	1.19 (30.23)
2	B (13)		1.000-20UNEF	1.12 (28.5)	1.19 (30.23)
3			1.188-18UNEF	1.25 (31.8)	1.38 (35.1)
4	C (15)	3.69 (97.9)	1.000-20UNEF	1.12 (28.5)	1.19 (30.23)
5			1.188-18UNEF	1.25 (31.8)	1.38 (35.1)
6			1.438-18UNEF	1.43 (36.3)	1.63 (41.4)
7		3.81 (101.0)	1.750-18UNS	1.62 (41.2)	1.94 (49.3)
8	D (17)		1.438-18UNEF	1.43 (36.3)	1.63 (41.4)
9			1.750-18UNS	1.62 (41.2)	1.94 (49.3)
10			2.00-18UNS	1.81 (46.0)	2.19 (55.6)
11	E (19)		1.438-18UNEF	1.43 (36.3)	1.63 (41.4)
12			1.750-18UNS	1.62 (41.2)	1.94 (49.3)
13		4.19 (111.1)	2.00-18UNS	1.81 (46.0)	2.19 (55.6)
14	F (23)		1.750-18UNS	1.62 (41.2)	1.94 (49.3)
15			2.00-18UN	1.81 (46.0)	2.19 (55.6)
16			2.250-16UN	2.18 (55.4)	2.44 (62.0)
17	G (25)		1.750-18UNS	1.62 (41.2)	1.94 (49.3)
18			2.00-18UN	1.81 (46.0)	2.19 (55.6)
19			2.250-16UN	2.18 (55.4)	2.44 (62.0)
20		4.50 (119.4)	2.500-16UN	2.56 (65.0)	2.69 (68.3)
21	H (29)		2.250-16UN	2.18 (55.4)	2.44 (62.0)
22			2.500-16UN	2.56 (65.0)	2.69 (68.3)
23	J (33)	4.83 (128.1)	2.500-16UN	2.56 (65.0)	2.69 (68.3)
24			3.00-16UN	3.06 (77.8)	3.19 (81.0)
25	A (11)	3.50 (92.8)	1.188-18UNEF	1.25 (31.8)	1.38 (35.1)

NOTES:
1. Shell sizes are provided within parenthesis for information and are not a part of the designator.
2. Dimensions are in inches.
3. Metric equivalents are given for information only.
4. Dimensions apply after plating.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS

Dimensions and configurations: See figure 1.

Interface dimensions in accordance with MIL-DTL-28840.

Material: Aluminum alloy in accordance with, ASTM B85/B85M, ASTM B108/B108M, ASTM B26/B26M, ASTM B209, ASTM B211, or ASTM B221.

Corrosion resistant steel shall be in accordance with ASTM A276, ASTM A580/A580M or ASTM A473, class 316".

Elastomers, silicone rubber shall be in accordance with A-A-59588.

Finish: Aluminum components shall be plated in accordance with MIL-DTL-28840. Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivate in accordance with SAE-AMS-2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:

M28840 /27 16 W A

Military designator

Specification sheet number

Dash number

Material and finish designator

Style designator
A – EMI/ non-environmental
B – EMI/ environmental

W – Aluminum, cadmium plate olive drab over electroless nickel
D – Corrosion resistant steel Cadmium plate, black
F – Wrought aluminum, pure electrodeposited aluminum – non-reflective
L - Wrought aluminum, Nickel Fluorocarbon Polymer non-reflective
S - Wrought aluminum, zinc nickel – non-reflective

Qualification: Two samples in each shell size for which qualification is desired shall be tested in accordance MIL-DTL-28840.

Qualification of MIL-DTL-28840/9 will qualify MIL-DTL-28840/27 by similarity for group 5 tests.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Referenced documents. In addition to MIL DTL-28840, this document references the following:

A-A-59588	ASTM B85/B85M	SAE-AMS-2700
MIL-DTL-28840/4	ASTM B26/B26M	
MIL-DTL-28840/9	ASTM B108/B108M	
MIL-DTL-28840/22	ASTM B209	
ASTM A276	ASTM B211	
ASTM A580/A580M	ASTM B221	
ASTM A473		

CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2008-012)

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